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A Cramér–von Mises statistic for randomly censored data

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Summary

The asymptotic distributions of Cramér–von Mises type statistics based on the product-limit estimate of the distribution function of a certain class of randomly censored observations are derived; the asymptotic significance points of the statistics for various degrees of censoring are given. The statistics are also partitioned into orthogonal components in the manner of Durbin & Knott (1972). The asymptotic powers of the statistics and their components against normal mean and variance shifts, exponential scale shifts, and Weibull alternatives to exponentiality are compared. Data arising in a competing risk situation are examined, using the Cramér–von Mises statistic.

Some key words: Censored data; Competing risk; Cramér–von Mises statistic; Product-limit estimate.

1. Introduction

An important problem in the analysis of survival data is whether the observed survival pattern for a cohort of individuals can be closely explained by a particular mathematical model with readily interpretable characteristics. Because the data arising from survival studies are often censored, because of deaths from competing causes or failure to observe individuals until time of death, statistical procedures formally relying upon the empirical distribution function can instead be defined in terms of the standard life table, or product-limit, estimate of the survival distribution. Our aim in this paper is to examine a Cramér–von Mises type statistic for tests of goodness of fit, based on the product-limit empirical distribution function, when the data are subject to random censorship. In § 2 we derive its asymptotic distribution under particular models of censorship; we investigate also the asymptotic properties of the components of the Cramér–von Mises statistic, which have been shown by Durbin & Knott (1972) to be of particular value in testing for specific aspects of the data. In § 3 we study briefly the asymptotic power of this Cramér–von Mises statistic and its components against normal mean shift and variance shift alternatives, exponential scale shifts, and Weibull alternatives to exponentiality. In § 4 we examine survival data with competing risks.

We have derived the asymptotic distribution of a Cramér–von Mises type statistic only for a particular distributional form of random censorship; nevertheless, we believe that this pattern of censorship often does occur, and we are confident that our methods can be extended to other random censorship patterns.

2. Distribution of the test statistic

Let $X_1^0, \ldots, X_n^0$ be independent random variables having a common continuous distribution function F^0. We may characterize a model of random censorship on the X_j^0 by means of censoring variables $T_1, \ldots, T_n$, drawn independently of the X_j^0 from a continuous distribution H. The X_j^0 are censored on the right by the T_j so that the observations available to us consist of the pairs (X_j, δ_j) $(1 \leqslant j \leqslant n)$, where $X_j = \min(X_j^0, T_j)$, and $\delta_j = 1$ or 0 as $X_j = X_j^0$ or T_j. Hence the observed X_j constitute a random sample from the distribution function F which satisfies $(1-F) = (1-F^0)(1-H)$.

Suppose that we wish to test the null hypothesis

$$H_0 : F^0 = F^*, \tag{2·1}$$

where F^* is a specified continuous distribution function. Had none of the X_j^0 been censored, we could have tested (2·1) with a Cramér–von Mises statistic using the empirical distribution function of the observations. With censoring, full knowledge of the empirical distribution is unavailable; we can, nevertheless, calculate the product-limit estimate $\hat{F}_n^0$ of F^0, which is given by

$$1 - \hat{F}_n^0(t) = \begin{cases} \prod_{m:X_m \leqslant t} \{(n-R_m)/(n-R_m+1)\}^{\delta_m} & (t < X_{(n)}), \\ 0 & (t \geqslant X_{(n)}), \end{cases}$$

where $X_{(n)} = \max(X_1, \ldots, X_n)$ and R_m is the rank of $(X_m, 1-\delta_m)$ in the lexicographic ordering of the sequence $(X_1, 1-\delta_1), \ldots, (X_n, 1-\delta_n)$. The product-limit estimate has been studied extensively by Kaplan & Meier (1958), Efron (1967), Meier (1975), and Breslow & Crowley (1974); under fairly general conditions the product-limit estimate is consistent and the random process determined by it converges weakly to a Gaussian process. In particular, we have the following result (Breslow & Crowley, 1974, Theorem 5).

THEOREM 2·1. *Suppose that F^0 is continuous, and that $(1-H) = (1-F^0)^\beta$, with β a positive constant. Then for $0 < t < T$, T being any finite value such that $F(T) < 1$, the random function $Y_n(t) = n^{\frac{1}{2}}\{\hat{F}_n^0(t) - F^0(t)\}$ converges weakly to a Gaussian process $Y(t)$ with*

$$E\{Y(t)\} = 0, \tag{2·2}$$

$$\operatorname{cov}\{Y(s), Y(t)\} = \{1-F^0(s)\}\{1-F^0(t)\}\int_0^s (1-F^0)^{-(2+\beta)} dF^0 \quad (s \leqslant t). \tag{2·3}$$

The constant β can be interpreted as the 'censoring parameter': $\beta = 0$ corresponds to no censoring, and the expected proportion of censored observations increases with β.

We assume throughout the remainder of this section that F^0 and H are related as in Theorem 2·1. We utilize the theorem to derive the asymptotic distribution of a Cramér–von Mises type statistic for testing (2·1), based on $\hat{F}_n^0$. By taking the probability integral transformation, (2·1) is equivalent to the hypothesis that $X_1^0, \ldots, X_n^0$ are uniformly distributed on (0, 1). We introduce the statistic

$$\psi_n^2 = \int_0^1 Y_n^2(t)\, dt = n\int_0^1 \{\hat{F}_n^0(t) - t\}^2\, dt$$

as a measure of the discrepancy or 'distance' between $\hat{F}_n^0$ and the uniform (0, 1) distribution. Note that, in the case of no censoring, ψ_n^2 simplifies to W_n^2, the usual Cramér–von Mises statistic.

Let

$$\psi^2(t) = \int_0^1 Y^2(t)\, dt,$$

where $\{Y(.)\}$ is the Gaussian process given in Theorem 2·1, and define a sequence of measurable functions on the functionals by

$$A_m(\psi_n^2) = \int_0^{1-1/m} Y_n^2(t)\, dt, \quad A_m(\psi^2) = \int_0^{1-1/m} Y^2(t)\, dt.$$

We may readily verify the conditions of Proposition 1 of Wichura (1971, p. 1769) to conclude that the asymptotic distribution of ψ_n^2 is the same as that of ψ^2.

We now obtain the Karhunen–Loéve expansion of the process $\{Y(.)\}$ (Ash & Gardner, 1975, §1·4). In particular, when F^0 is the uniform (0, 1) distribution function, the covariance kernel (2·3) of $\{Y(.)\}$ becomes

$$K(s,t) = (1+\beta)^{-1}(1-s)(1-t)[\{1-\min(s,t)\}^{-(1+\beta)}-1] \quad (0 \leqslant s,t < 1). \tag{2·4}$$

We note that K is not necessarily continuous at (1, 1) nor is it integrable for $\beta \geqslant 2$. Nevertheless, for $\beta < 2$ we have (Hammerstein, 1927; Anderson & Darling, 1952, p. 198) that in every domain in the interior of the unit square, K can be expanded in the uniformly convergent series

$$K(s,t) = \sum_{j=1}^{\infty} \lambda_j \phi_j(s)\phi_j(t);$$

in this representation, λ_j is an eigenvalue and ϕ_j the corresponding orthonormal eigenfunction of the integral equation $\int K(s,t)\phi_j(s)\,ds = \lambda_j\phi_j(t)$. Letting $X_1, X_2, \ldots$ be independent, identically distributed $N(0,1)$ random variables, we may define

$$Z(t) = \sum_{j=1}^{\infty} (\lambda_j)^{\frac{1}{2}} \phi_j(t) X_j;$$

the series converges in the mean and with probability one for every t ($\epsilon \leqslant t \leqslant 1-\epsilon$), and $Z(t)$ determines the same process as $Y(t)$ in this interval. By appropriate modification of the Kac–Siegert (1947) argument, it is not difficult to show (Anderson & Darling, 1952; Hájek & Sidák, 1967, p. 192) that, in distribution,

$$\psi^2 = \int_0^1 Y^2(t)\,dt = \int_0^1 Z^2(t)\,dt = \sum_{j=1}^{\infty} \lambda_j X_j^2.$$

Also ψ_n^2 can be decomposed into a set of asymptotically independent components (Durbin & Knott, 1972) which may be of greater use in distinguishing specific aspects of the data than the overall statistic ψ_n^2. Define

$$X_{jn} = (\lambda_j)^{-\frac{1}{2}} \int_0^1 Y_n(t)\phi_j(t)\,dt, \quad X_j = (\lambda_j)^{-\frac{1}{2}} \int_0^1 Y(t)\phi_j(t)\,dt \quad (j = 1, 2, \ldots).$$

Since

$$Y_n(t) = \sum_{j=1}^{\infty} (\lambda_j)^{\frac{1}{2}} \phi_j(t) X_{jn},$$

it follows immediately that

$$\psi_n^2 = \int_0^1 Y_n^2(t)\,dt = \sum_{j=1}^{\infty} \lambda_j X_{jn}^2.$$

Upon defining a sequence of measurable functions by

$$A_m(X_{jn}) = (\lambda_j)^{-\frac{1}{2}} \int_0^{1-1/m} Y_n(t)\phi_j(t)\,dt, \quad A_m(X_j) = (\lambda_j)^{-\frac{1}{2}} \int_0^{1-1/m} Y(t)\phi_j(t)\,dt,$$

we may conclude, using Proposition 1 of Wichura (1971), that X_{jn} converges in distribution to X_j for each j. Following Durbin & Knott, we may base tests of (2·1) on the components $\{X_{jn}\}$, which are asymptotically independent normally distributed with zero mean and unit variance.

We turn now to the decomposition of the kernel (2·3) in terms of its eigenvalues and eigenvectors. This is a rather involved procedure, the details of which we relegate to

Appendix 1. The eigenvalues are zeros of a Bessel function of order depending on β: for given β, $\lambda_j = \{2/(2-\beta)\}^2(1/\xi_j^2)$, where ξ_j is the jth zero of a Bessel function of the first kind and of order $(1+\beta)/(2-\beta)$. We can also show (Anderson & Darling, 1952, equation (4·24)) that

$$E(\psi^2) = \frac{1}{3(2-\beta)}, \quad \operatorname{var}(\psi^2) = \frac{2}{(1+\beta)^2}\left\{\frac{\beta-1}{(2-\beta)(5-\beta)}+\frac{1}{9}\right\}.$$

To obtain the distribution of ψ^2, we approximate it by

$$\bar{\psi}^2 = \sum_{j=1}^{m} \lambda_j X_j^2 + \lambda X^2, \tag{2·5}$$

where X^2 is a χ^2 variable with ν degrees of freedom, λ and ν being chosen so that ψ^2 and $\bar{\psi}^2$ have the same mean and variance. The characteristic function of $\bar{\psi}^2$ is

$$E\{\exp(it\bar{\psi}^2)\} = \prod_{j=1}^{m} (1-2i\lambda_j t)^{-\frac{1}{2}}(1-2i\lambda_j t)^{-\frac{1}{2}\nu},$$

a form that can be inverted numerically using Imhof's method (Imhof, 1961); upon doing so, we may then obtain percentage points of the distribution of $\bar{\psi}^2$ with an inverse interpolation scheme. This technique has been found to give excellent results in quite similar situations (Durbin & Knott, 1972; Pettitt & Stephens, 1976). We allowed m to be 20 in our calculations, and used an adaptive procedure in performing the numerical integrations. The calculations were performed on the National Institutes of Health PDP-10 computer. The percentage points so obtained for the particular cases $\beta = \frac{1}{2}$, 1, $\frac{3}{2}$ are given in Table 1, with the percentage points for W^2 taken from Anderson & Darling (1952). In addition, in Fig. 1 are plotted the cumulative distribution functions of W^2 and ψ^2 for $\beta = \frac{1}{2}$, 1, $\frac{3}{2}$. Note that the distributions become more skew as the amount of censoring increases; in hypothesis testing situations the use of the critical values for W^2 with the statistic ψ^2 can result in substantial departure from the nominal level.

3. Power considerations

We investigate in this section the asymptotic properties of tests of (2·1) based on the statistics ψ_n^2 and their individual components. Let $F^0(x) = F^0(x, \theta)$, where the parameter θ specifies the distribution completely; further, let the censoring distribution H be related to F^0 as in Theorem 2·1. In terms of θ, we may reformulate the null hypothesis (2·1) as

$$H_0 : \theta = \theta_0. \tag{3·1}$$

We set $t = F^0(x, \theta_0)$ and $Y_n(t) = n^{\frac{1}{2}}\{\hat{F}_n^0(t) - t\}$; then, under the sequence of alternatives

$$H_{1n} : \theta = \theta_{1n} = \theta_0 + \gamma n^{-\frac{1}{2}}, \quad E\{Y_n(t)\} = n^{\frac{1}{2}}\{F^0(x, \theta_1) - F^0(x, \theta_0)\} + o_p(1), \tag{3·2}$$
$$\operatorname{cov}\{Y_n(s), Y_n(t)\} = n \operatorname{cov}\{\hat{F}_n^0(s), \hat{F}_n^0(t)\}.$$

We may also show by arguments quite similar to those given in the null case that the process $\{Y_n(.)\}$ under H_{1n} converges to a Gaussian process $\{Y(.)\}$ with $E\{Y(t)\} = \gamma' g(t)$, where $g(t) = [\partial F(x, \theta)/\partial\theta]_{\theta=\theta_0}$, and covariance kernel given by (2·4), the same as the null case. Furthermore, under (3·2) the X_{jn} are asymptotically independent with means $\gamma'\eta_j$ and variance 1, where

$$\eta_j = \lambda_j^{-\frac{1}{2}} \int_0^1 \phi_j(t)\, g(t)\, dt. \tag{3·3}$$

Table 1. *Asymptotic percentage points of ψ_n^2 for censoring parameters* $\beta = 0, \frac{1}{2}, 1, \frac{3}{2}$

Cumulative probability	$\beta = 0$	$\beta = \frac{1}{2}$	$\beta = 1$	$\beta = \frac{3}{2}$
0·1	0·0460	0·0702	0·1254	0·3231
0·2	0·0622	0·0926	0·1598	0·3909
0·3	0·0786	0·1145	0·1924	0·4513
0·4	0·0970	0·1385	0·2268	0·5125
0·5	0·1189	0·1666	0·2660	0·5792
0·6	0·1466	0·2014	0·3134	0·6565
0·7	0·1843	0·2478	0·3749	0·7530
0·8	0·2412	0·3166	0·4639	0·8870
0·9	0·3473	0·4434	0·6241	1·1176
0·95	0·4614	0·5791	0·7936	1·354
0·99	0·7435	0·9155	1·213	1·926
0·995	0·8694	1·066	1·401	2·18
0·999	1·168	1·42	1·85	2·79

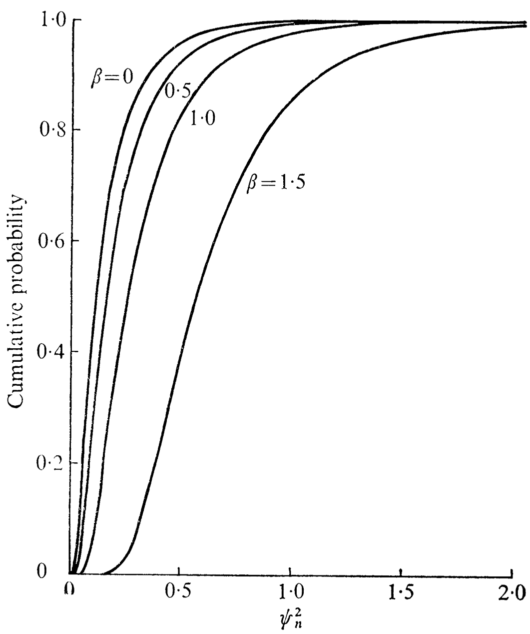

Fig. 1. Asymptotic cumulative distribution function of ψ_n^2 for censoring parameters $\beta = 0, \frac{1}{2}, 1, \frac{3}{2}$.

As in § 2, the λ_j are the eigenvalues and the ϕ_j the corresponding eigenfunctions of the covariance kernel (2·4). The limiting distribution of ψ_n^2 is that of the infinite sum $\Sigma\lambda_j X_j^2$, the X_j being independent $N(\gamma'\eta_j, 1)$.

To demonstrate the relation of the degree of censoring to the limiting powers of tests based on the statistics ψ_n^2 and the individual components X_{jn}, we consider four cases:

(i) F^0 is normal with mean θ and variance 1;
(ii) F^0 is normal with mean 0 and variance θ;
(iii) F^0 is exponential with scale parameter θ;
(iv) F^0 is Weibull, that is $F^0(t) = 1 - \exp(-t^\theta)$.

For (i), we test $H_0:\theta = 0$ versus $H_{1n}:\theta = n^{-\frac{1}{2}}\gamma$; for (ii), (iii), and (iv), we test $H_0:\theta = 1$ versus $H_{1n}:\theta = 1+n^{-\frac{1}{2}}\gamma$. With each case, we make power comparisons for both the one-sided and two-sided situations, the tests being taken at the 5 % level. In each instance, two values of γ were selected so that the asymptotic power of the 'best' test was 0·5 and 0·95. The asymptotic power of the components under the various alternatives can be readily calculated, with the η_j of (3·3) being computed by numerical integration. The asymptotic power of ψ_n^2 for the various alternatives is approximated by pr $(\bar{\psi}^2 > c_\alpha)$; here, $\bar{\psi}^2$ is given as in (2·5), but with noncentral chi-squared random variables so chosen that $\bar{\psi}^2$ has the same mean and variance under H_0 and the same mean under H_1 as ψ^2, and c_α is the $(1-\alpha)$-level critical value of ψ^2 for the appropriate censoring parameter β chosen from Table 1. The probability is calculated using Imhof's technique as in § 2. The results are given in Table 2.

For tests against mean shift with normally distributed random variables, the first component is as powerful asymptotically as ψ_n^2, for any value of the censoring parameter β. However, as β increases, meaning that the proportion of censoring is increasing, the asymptotic powers of both X_{1n} and ψ_n^2 decrease. Although the other components have little power against normal mean shift alternatives, it is of interest to note that the asymptotic powers of X_{2n} and X_{4n} actually increase slightly as β increases. This is explained by the fact that the eigenfunctions of $K(s,t)$, sine functions if $\beta = 0$, are no longer symmetric functions if $\beta > 0$; hence the η_j, the asymptotic means of the X_{jn}, are no longer the inner products of symmetric and antisymmetric functions, and consequently zero, for positive β. For testing against variance shift with normally distributed random variables, only the second component X_{2n} is much more powerful asymptotically than ψ_n^2. Again we note patterns with power of X_{jn} as β increases: the asymptotic powers of X_{1n} and X_{4n} decrease, but that of X_{3n} tends to increase. For testing against scale shift with exponentially distributed random variables, the first component X_{1n} is asymptotically about as powerful as ψ_n^2; the asymptotic power of X_{1n} decreases, but the asymptotic powers of the other components tend to increase as β increases. For testing against Weibull alternatives to exponentially distributed random variables, the second component X_{2n} is substantially more powerful than ψ_n^2 up to $\beta = \frac{3}{2}$; the fourth component X_{4n} also compares favourably with ψ_n^2 in one-sided tests. Note that the trends in the asymptotic powers of the components as β increases are quite similar for normal mean shifts and exponential scale shifts, and for normal variance shifts and Weibull alternatives to exponentiality; and in both cases, certain components perform much better than the overall Cramér–von Mises statistic.

In Fig. 2 (a) and (b) are plotted the asymptotic powers of W_n^2 and ψ_n^2 for $\beta = \frac{1}{2}, 1, \frac{3}{2}$ against exponential scale shifts and against Weibull alternatives to exponentiality respectively, at alpha level 0·05. As in Table 2, the powers are against the local alternatives $\theta = 1+n^{-\frac{1}{2}}\gamma$, the null hypothesis being $\theta = 1$. Clearly, the Cramér–von Mises statistic is less sensitive to Weibull departures from exponentiality than to exponential scale shift. Also, although tests based on ψ_n^2 remain asymptotically unbiased, there is a substantial loss of efficiency attendant with increased censoring.

4. EXAMPLE

Among the conclusions of a clinical trial of treatments for prostatic cancer (V.A.C.U.R.G., 1967) was that oestrogen therapy was associated with an increased risk of cardiovascular disease, while the clinical benefits with respect to the cancer were smaller than was generally realized. In the trial, of 211 patients with inoperable cancer who were treated with oestrogen, 87, i.e. 41·2 %, died of cancer of the prostate, 102, i.e. 48·3 %, died of other causes, and 22, i.e. 10·4 %, were withdrawals. Prior experience suggested that had the patients not been

Table 2. *Asymptotic powers of components and ψ_n^2 against shifts in normal mean and variance and exponential scale, for censoring parameters $\beta = 0, \frac{1}{2}, 1, \frac{3}{2}$*

Test statistic	$\beta=0$ Power of best test 0·50	0·95	$\beta=\frac{1}{2}$ Power of best test 0·50	0·95	$\beta=1$ Power of best test 0·50	0·95	$\beta=\frac{3}{2}$ Power of best test 0·50	0·95
			(a) One-sided situation					
			(I) Normal variates, mean shift					
X_{1n}	0·466	0·930	0·412	0·885	0·344	0·800	0·252	0·621
X_{2n}	0·050	0·050	0·076	0·111	0·108	0·204	0·136	0·290
X_{3n}	0·106	0·197	0·096	0·166	0·095	0·165	0·106	0·197
X_{4n}	0·050	0·050	0·063	0·078	0·076	0·110	0·090	0·150
ψ_n^2	0·342	0·877	0·285	0·805	0·224	0·696	0·140	0·480
			(II) Normal variates, variance shift					
X_{1n}	0·050	0·050	0·044	0·039	0·039	0·029	0·033	0·021
X_{2n}	0·336	0·787	0·261	0·642	0·178	0·420	0·095	0·165
X_{3n}	0·050	0·050	0·074	0·107	0·097	0·171	0·094	0·161
X_{4n}	0·162	0·372	0·130	0·270	0·109	0·207	0·094	0·160
ψ_n^2	0·072	0·205	0·069	0·166	0·065	0·131	0·058	0·089
			(III) Exponential variates, scale shift					
X_{1n}	0·392	0·864	0·337	0·789	0·273	0·669	0·193	0·466
X_{2n}	0·131	0·276	0·150	0·335	0·161	0·367	0·150	0·333
X_{3n}	0·109	0·207	0·019	0·006	0·121	0·244	0·125	0·255
X_{4n}	0·086	0·137	0·093	0·160	0·102	0·185	0·109	0·208
ψ_n^2	0·284	0·806	0·235	0·723	0·184	0·607	0·116	0·391
			(IV) Exponential variates, Weibull alternatives					
X_{1n}	0·019	0·006	0·018	0·005	0·016	0·004	0·016	0·004
X_{2n}	0·459	0·925	0·318	0·758	0·180	0·428	0·075	0·110
X_{3n}	0·072	0·102	0·105	0·193	0·120	0·242	0·092	0·154
X_{4n}	0·213	0·521	0·166	0·384	0·134	0·285	0·099	0·177
ψ_n^2	0·116	0·486	0·108	0·387	0·098	0·297	0·083	0·207
			(b) Two-sided situation					
			(I) Normal variates, mean shift					
X_{1n}	0·460	0·928	0·395	0·876	0·316	0·778	0·214	0·571
X_{2n}	0·050	0·050	0·057	0·075	0·078	0·146	0·100	0·223
X_{3n}	0·076	0·140	0·069	0·115	0·068	0·114	0·076	0·140
X_{4n}	0·050	0·050	0·052	0·057	0·057	0·075	0·065	0·102
ψ_n^2	0·457	0·928	0·383	0·871	0·302	0·777	0·185	0·564
			(II) Normal variates, variance shift					
X_{1n}	0·050	0·050	0·051	0·052	0·052	0·058	0·056	0·071
X_{2n}	0·307	0·763	0·223	0·595	0·138	0·352	0·069	0·114
X_{3n}	0·050	0·050	0·057	0·073	0·070	0·118	0·068	0·111
X_{4n}	0·123	0·303	0·094	0·205	0·078	0·149	0·067	0·110
ψ_n^2	0·085	0·258	0·078	0·202	0·072	0·154	0·062	0·099
			(III) Exponential variates, scale shift					
X_{1n}	0·372	0·852	0·309	0·766	0·236	0·626	0·153	0·401
X_{2n}	0·096	0·210	0·112	0·267	0·122	0·299	0·112	0·265
X_{3n}	0·078	0·148	0·082	0·160	0·088	0·181	0·090	0·191
X_{4n}	0·063	0·093	0·067	0·110	0·073	0·130	0·078	0·149
ψ_n^2	0·382	0·872	0·317	0·802	0·246	0·694	0·149	0·468
			(IV) Exponential variates, Weibull alternatives					
X_{1n}	0·080	0·153	0·085	0·171	0·090	0·188	0·090	0·190
X_{2n}	0·452	0·922	0·287	0·729	0·141	0·361	0·057	0·074
X_{3n}	0·056	0·069	0·075	0·136	0·087	0·179	0·066	0·106
X_{4n}	0·173	0·461	0·127	0·316	0·098	0·218	0·071	0·124
ψ_n^2	0·155	0·594	0·139	0·473	0·121	0·358	0·099	0·245

treated with oestrogen, their deaths from cancer of the prostate would be exponentially distributed with mean 100 months. To test whether the survival experience among the oestrogen-treated patients, considering only deaths from prostatic cancer, was consistent with this exponential distribution, we calculated the estimated survival curve for prostatic cancer deaths using the product-limit method, treating deaths due to other causes as well as withdrawals as censored observations. The Cramér–von Mises statistic, ψ_n^2, was 0·484. In Appendix 2 we discuss briefly the computing of ψ_n^2.

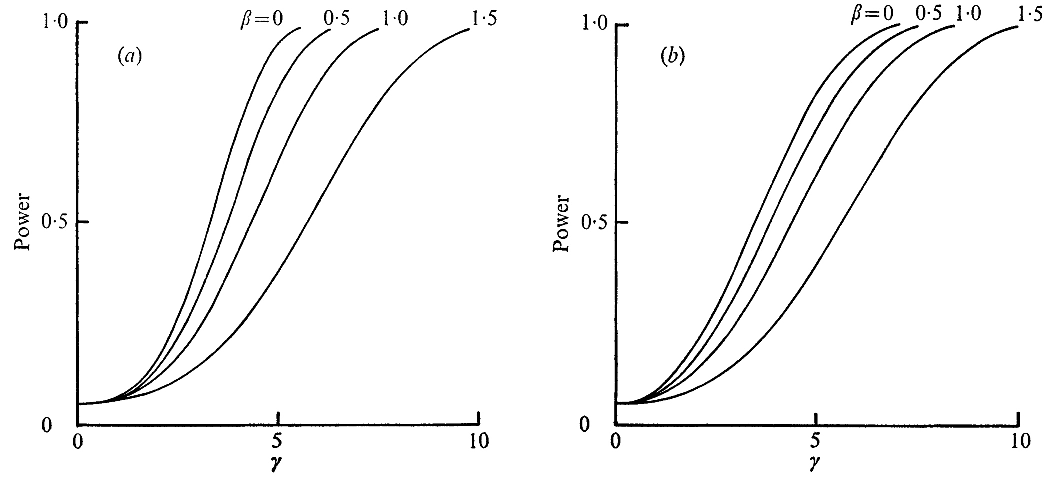

Fig. 2. Asymptotic power curves of tests based on ψ_n^2 at $\alpha = 0{\cdot}05$, for censoring parameters $\beta = 0, \frac{1}{2}, 1, \frac{3}{2}$; (a) against exponential scale shifts, (b) against Weibull alternatives to exponentially distributed random variates.

Since by far the principal cause of death among the noncancer deaths was cardiovascular disease, it was assumed *a priori* that the censoring distribution was exponential; that is, that $(1-H) = (1-F^0)^\beta$. The censoring parameter β may be consistently estimated from the sample, for the observed proportion of censored observations, here 0·588, converges to its expected value, $\beta/(\beta+1)$. As ψ_n^2 is a function solely of the empirical distribution function of the uncensored observations and hence has a value not dependent on the parameter β, we may use the estimate of β to determine the appropriate critical value. Thus, using $\beta = \frac{3}{2}$ and comparing $\psi_n^2 = 0{\cdot}484$ with the critical values in Table 1, we conclude that the pattern of cancer deaths among the oestrogen treated patients is consistent with the hypothesized distribution.

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APPENDIX 1

Determination of the orthogonal decomposition of the covariance kernel

Suppose that λ is an eigenvalue and ϕ the corresponding eigenfunction of the covariance kernel $K(s,t)$ of (2·4). We assume throughout that $0 < \beta < 2$. We may show by standard techniques that ϕ satisfies the differential equation

$$x^2 d^2\phi/dx^2 + \beta x\, d\phi/dx + \{x^{2-\beta}(1+\beta)/\lambda - \beta\}\phi = 0.$$

This equation is transformed to a Bessel equation by a change of variables $\Phi = x^{-p}\phi$ and $X = ax^q$, where

$$p = \tfrac{1}{2} - \tfrac{1}{2}\beta, \quad q = 1 - \tfrac{1}{2}\beta, \quad a = 2(1+\beta)^{\frac{1}{2}}\lambda^{-\frac{1}{2}}(2-\beta)^{-1},$$
$$\nu = \{(1-\beta)^2 + 4\beta\}^{\frac{1}{2}}(2-\beta)^{-1} = (1+\beta)/(2-\beta).$$

The general solution is (Jahnke & Emde, 1943, § VIII.7)

$$\phi(x) = x^p\{c_1 J_\nu(ax^q) + c_2 K_\nu(ax^q)\},$$

where J denotes a Bessel function of the first kind, and K a modified Bessel function. The bounded eigenfunctions are therefore

$$\phi_\lambda(x) = c_\lambda x^{\frac{1}{2}-\frac{1}{2}\beta} J_\nu\{2(1+\beta)^{\frac{1}{2}}\lambda^{-\frac{1}{2}}(2-\beta)^{-1}x^{1-\frac{1}{2}\beta}\},$$

and the eigenvalues can be determined from the positive real solution of

$$J_\nu\{2(1+\beta)^{\frac{1}{2}}\lambda^{-\frac{1}{2}}(2-\beta)^{-1}\} = 0.$$

The boundedness of the ϕ_λ follows immediately because $x^{\frac{1}{2}-\frac{1}{2}\beta}J_\nu\{x^{1-\frac{1}{2}\beta}\}$ behaves as $x^{\frac{1}{2}-\frac{1}{2}\beta}x^{\nu(1-\frac{1}{2}\beta)}$ near the origin (Abramowitz & Stegun, 1964, equation (9·1·10)). For integral ν, the roots of J_ν are tabulated by Abramowitz & Stegun (Table 9·5); the eigenfunctions may be integrated numerically to determine the normalizing constants c_λ.

Appendix 2

On computing formulae for ψ_n^2

If $\{Y^0_{(j)}\}$ $(1 \leqslant j \leqslant n)$ constitute the ordered uncensored observations on (0, 1) obtained by the probability integral transformation $Y^0_{(j)} = F^0(X^0_{(j)})$, then, adopting the convention $Y^0_{(0)} = 0$, $Y^0_{(n+1)} = 1$, we have

$$\begin{aligned}\psi_n^2 &= \int_0^1 \{\hat{F}^0_n(t) - t\}^2 dt = n\sum_{j=1}^{n+1}\int_{Y_{(j-1)}}^{Y_{(j)}} \{\hat{F}^0_n(Y_{(j-1)}) - t\}^2 dt \\ &= n\sum_{j=1}^{n+1}[\{\hat{F}^0_n(Y_{(j-1)})\}^2(Y_{(j)} - Y_{(j-1)}) - \{\hat{F}^0_n(Y_{(j-1)})\}(Y^2_{(j)} - Y^2_{(j-1)})] + \tfrac{1}{3}n \\ &= n\sum_{j=1}^{n+1}\{\hat{F}^0_n(Y_{(j-1)})\}(Y_{(j)} - Y_{(j-1)})\{\hat{F}^0_n(Y_{(j-1)}) - (Y_{(j)} + Y_{(j-1)})\} + \tfrac{1}{3}n.\end{aligned}$$

This last formula or variants thereof is easily implemented.

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